Roll No. .....

BBA–502

B. B. A. (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

QUANTITATIVE TECHNIQUES IN MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define Operations Research. Explain the following terminology : (CO1)

(i) Objective Function

(ii) Constraints

(iii) Slack variables

(iv) Optimum solution

OR

(b) The manager of an oil refinery must decide on the optimal mix of 2 possible blending processes of which the inputs and outputs per production run are as follows : (CO1)

Process	Input		Output	
	Crude A	Crude B	Gasoline X	Gasoline Y
1	6	3	6	9
2	5	6	5	5

P. T. O.

The maximum availability of crude A and B is 250 units and 200 units respectively. The market requirement shows that at least 150 units of gasoline X and 130 units of gasoline Y must be produced. The profits per production run from processes 1 and 2 are ₹ 40 and ₹ 50 respectively. Formulate the problem for maximizing the profit.

2. (a) What is LPP ? Discuss the steps used in LPP to solve the problem. (CO1)

OR

(b) Solve the following LPP using Graphical Method : (CO1)

Max. :

$$Z = 2x_1 + 5x_2$$

Subject to :

$$x_1 + 4x_2 \leq 24$$

$$3x_1 + x_2 \leq 21$$

$$x_1 + x_2 \leq 9$$

where $x_1, x_2 \geq 0$.

3. (a) A firm manufactures 3 products A, B and C. The profits are ₹ 3, ₹ 2 and ₹ 4 respectively. The firm has 2 machines and below is the required processing time in minutes for each machine on each product. Machine C and D has 2,000 and 2,500 machine respectively. The firm must manufacture 100 A's, 200 B's and 50 C's but not more than 150 A's. (CO2)

	Products		
	A	B	C
Machine C	4	3	5
Machine D	2	2	4

OR

(b) Solve the following transportation problem using the NWCM method. (CO3)

	A	B	C	Production at plant
X	11	21	16	14
Y	7	17	13	26
Z	11	23	21	36
Requirement	18	28	25	

4. (a) Solve the following LPP using Simplex Method : (CO2)

Max. :

$$Z = 10x_1 + 5x_2$$

Subject to :

$$4x_1 + 5x_2 \leq 100$$

$$5x_1 + 2x_2 \leq 80$$

where $x_1, x_2 \geq 0.$

OR

(b) Solve the following LPP using Big-M method : (CO2)

Min. :

$$Z = 5x_1 + 6x_2$$

Subject to :

$$2x_1 + 5x_2 \geq 1500$$

$$3x_1 + x_2 \geq 1200$$

where $x_1, x_2 \geq 0.$

5. (a) Solve the following transportation problem using Matrix Minima method : (CO3)

	D1	D2	D3	D4	D5	Availability
W1	3	4	6	8	8	20
W2	2	10	0	5	8	30
W3	7	11	20	40	3	15
W4	1	0	9	14	16	13
Requirement	40	6	8	18	6	78

OR

(b) Solve the following transportation problem using VAM method : (CO3)

	A	B	C	D	Availability
I	21	16	25	13	11
II	17	18	14	23	13
III	32	27	18	41	19
Requirement	6	10	12	15	43

450